

1. The ability of one cytokine to act on several cell is called?
 - a. Redundancy
 - b. synergism
 - c. Pleiotropic
 - d. additive

2. Perforins and granzymes belong to the following cell?
 - a. Th2
 - b. TH1
 - c. macrophages
 - d. CD8+ T cells

3. The following participate in intracellular activation of T cells.
 - a. phospholipase γ
 - b. NF-AT transcription factor
 - c. All of the choices are correct.
 - d. ZAP-70.
 - e. ITAM

4. Ag-specific lymphocytes are only activated in lymph nodes because they have specific?
 - a. cytokine Receptor
 - b. Hormone receptor
 - c. interleukin
 - d. enzyme substrate

5. The following is a cytotoxic CTL killing mechanism against viral-infected cells?
 - a. IFN-gamma secretion
 - b. Fas-FasL.
 - c. ICOS-ICOSL
 - d. IL-17 secretion

6. Eating strawberry cause urticaria due to the following?

- a. Monovalent haptens present in strawberry
- b. Lectins that cross-link surface IgE**
- c. Autoantibodies against Fcε R1
- d. Anaphylatoxins
- e. Anti-IgE autoantibodies

7. A patient with rheumatic fever develops a sore throat from which beta-hemolytic streptococci are cultured, The patient is started on treatment with penicillin, and the sore throat resolves within several days, However, 7 days after initiation of penicillin therapy the patient develops a fever of 39°C, a generalized rash, and proteinuria, This MOST probably resulted from?

- a. an IgG-IgM response to penicillin.**
- b. a delayed hypersensitivity reaction to penicillin.
- c. a different infectious disease.
- d. recurrence of the rheumatic fever.
- e. an IgE response to penicillin.

8. The following autoimmune disease cause hypothyroidism?

- a. Systemic lupus erythematosus
- b. Myasthenia graves
- c. Graves disease
- d. rheumatoid arthritis
- e. Hashimoto's thyroiditis**

9. C3a and C5a anaphylatoxins?

- a. phagocytosis of IgE-coated bacteria.
- b. cause bacterial lysis.
- c. induce smooth muscle contraction.**
- d. aggregation of C4 and C2.
- e. enhance antibody production.

10. The most important function of the complex formed by complement components C5b,6,7,8,9 is?

- a. To enhance antibody production
- b. To perforate bacterial cell membranes**
- c. To opsonize viruses
- d. To release histamine from mast cells
- e. To inhibit immune complex formation

11. The function of Plasmacytoid dendritic cells is?

- a. inhibition of immune response
- b. secretion of IFN-alpha and beta**
- c. induction of plasma cells
- d. induction of T cells

12. An individual is called sensitized when his?

- a. mast cells have cross-linking surface IgE with allergen.
- b. mast cells have surface Fc epsilon R.
- c. mast cells release histamine.
- d. mast cells have surface bound IgE**

13. Down regulation of..... allow naive T cells to enter lymph nodes while upregulates.....allow them to migrate out lymph nodes?

- a. CD40, CD40L
- b. TCR, MHC
- c. B7, CD28
- d. L-selectin, integrin**

14. Expression of CD25 and CTLA-4 is a characteristic of?

- a. TH17
- b. TH1
- c. Treg
- d. TH2

15. Which one of the following is the most correct regarding the complement system?

- a. To perforate bacterial cell membranes
- b. To enhance antibody production
- c. To inhibit immune complex formation
- d. To opsonize viruses
- e. To release histamine from mast cells

16. As a response to extracellular pathogens, the following cell gets activated?

- a. TH1
- b. TH17
- c. TH2
- d. TH3

17. One of the following does not describe myasthenia graves?

- a. An autoimmune disease
- b. removal of thymus leads to regression of the disease
- c. cause muscle weakness
- d. caused by antibodies against acetylcholine receptor
- e. cause hyperthyroidism

18. Cytokines that stimulate hematopoiesis include?

- a. CTLA-4
- b. G-CSF, M-CSF
- c. IL-3, IL-4, IL-5
- d. IL-6 , TNF

19. The following characterize the immune response against HIV EXCEPT?

- a. production of antimicrobial peptides (defensins)
- b. The early response to HIV infection serves to clear most of the virus present in the blood
- c. AIDS patients produce small amounts of antibodies.
- d. The most immunogenic HIV molecules that elicit Ab responses are gp120 and gp41

20. The following are proinflammatory cytokines?

- a. IL-3, IL-4, IL-5
- b. CTLA-4
- c. G-CSF, M-CSF
- d. IL-6 , TNF

21. The following does not characterize mature dendritic cells?

- a. secretion of IL-12
- b. MHC synthesis
- c. B7 expression
- d. CD28 surface expression

22. The classic complement pathway is initiated by interaction of C1 with one of the following?

- a. mannose-binding lectin
- b. factor B.
- c. bacterial lipopolysaccharides.
- d. antigen.
- e. antigen-IgG complexes.

23. The following describe chemokines EXCEPT:

- a. potent attractors of antibodies
- b. involve IL-8 and RANTES
- c. belong to innate immunity.
- d. low molecular weight proteins
- e. Have Cysteine-rich conserved sequences

24. The following costimulatory pairs participate in the immunological synapse of APC-T cells?

- a. IL -2 ----- IL -2 receptor
- b. MHC ----- TCR
- c. B7----- CD40
- d. ICOSL ----- ICOS

25. The following cells produce granzyme inhibitor to protect dendritic cells (DCs) from granzymes of CD8+ T cells ?

- a. Both DCs and CTL
- b. memory TH
- c. memory CTL cells
- d. DCs

26. The following event does not occur in the germinal centers of activated B cells?

- a. Somatic DNA recombination
- b. plasma cell formation
- c. somatic hypermutation
- d. formation of memory B cells
- e. class switching

28. The following describe Platelet-activating factor EXCEPT?

- a. derived from phospholipids
- b. cause aggregation of platelets and release content of serotonin
- c. Highly potent
- d. a stored in the granules inside mast cells
- e. cause bronchconstriction and vasodilation

29. A patient with severe asthma gets no relief from antihistamines. The symptoms are MOST likely to be caused by?

- a. Proteases
- b. interleukin-2.
- c. bradykinin.
- d. leukotrienes.
- e. serotonin.

30. The failure to treat AIDS might be due to the following EXCEPT?

- a. The virus covers itself with CD59 and DAF to be protected from complement
- b. lack of fidelity of reverse transcriptase and RNA polymerase
- c. Activated lymphocytes have inactive cyt deaminase
- d. Inhibition of IFN-alpha that prevent viral infection
- e. The virus is able to produce a glycan shield

31. Naïve T cells resist HIV infection because they have active?

- a. Cyt deaminase enzyme
- b. Uridine deaminase enzyme
- c. Reverse transcriptase
- d. protease
- e. HGPR transferase

33. One of the following does not characterize cytokines?

- a. Short half-life
- b. Secretion is brief
- c. stored as preformed mediators
- d. Unstable mRNA

34. Innocent bystander self-cells are protected from complement system by?

- a. binding to antibodies
- b. having surface human serum albumin
- c. having factor H protein that breakdown C5 convertase
- d. having surface CD59 protein that inhibit MAC formation

35. The following describe the killing process of tumor cells by CTL CD8+ cells EXCEPT?

- a. Cell to cell contact is necessary
- b. The killing is physiological and the target cell can regenerate again
- c. involve sequential activation of caspases proteolytic enzymes
- d. The killing is logarithmic
- e. Killing by apoptosis prevent the spread of the virus into other cells

36. HIV associated immune deficiency is a result of?

- a. All of the choices are correct
- b. Hypogammaglobulinemia
- c. Autoantibodies directed to medullary stromal cells
- d. destruction of the CD8 memory cell pool during primary infection
- e. Early destruction of the CD4+ T cells in the gut

37. A man was stung by a bee and had no reaction, The second time he was stung, he had local swelling immediately. There was no reaction the first time because of which of the following?

- a. He had no preformed IgE molecules specific for bee venom.
- b. He was tolerant to bee venom.
- c. He had no mast cells.
- d. The bee venom inhibit the specific IgE antibodies.
- e. He had blocking IgG antibodies to bee venom.

38. Features of T-independent immune response include?

- a. production of IgG and IgM
- b. involve immunological synapse
- c. occurs in people that lack a functional thymus
- d. give rise to memory cells.

39. Perforins and granzymes are synthesized by the following cell?

- a. Th2
- b. CD8+ T cells
- c. TH1
- d. macrophages

40. As a response to parasitic worms, the following cell gets activated?

- a. TH2
- b. TH17
- c. TH3
- d. TH1

41. Rheumatoid arthritis can be induced in rats by injection of?

- a. TNF- α inhibitor
- b. Collagen II
- c. Freund's incomplete adjuvant
- d. Anti-IL-6.
- e. Pooled normal rat IgG

42. The following is not a characteristic of follicular TH cells?

- a. release of IL-21
- b. Germinal center marker (CXCR5)
- c. Transcription factors Stat-1 and T-bet
- d. Transcription factor Bcl-6

43. The following describe the immunological status of a chronic latent phase of HIV infection?

- a. Integrated viral DNA remain as nonantigenic provirus
- b. The patient serum has no antibodies against HIV
- c. The ratio of CD4+/CD8+ = 2/1
- d. Number of viral-infected T cells in peripheral blood is high
- e. Drastic decline in CD8+

44. Superantigens cause a burst of cytokine production by T cells due to their ability to

- a. cross-link the VB segments of T-cell receptors with class II MHC molecules on APCs
- b. cross-link the Va segments of T-cell receptors with class II MHC molecules on APCs
- c. cross-link T-cell receptors and CD3
- d. cross-link multiple cytokine receptors on a large population of T cells
- e. cross-link CD3

45. IL-1, IL-6, and TNF- α are proinflammatory cytokines that are known

- a. cause increased vascular permeability
- b. act in concert with chemokines to promote migration of inflammatory cells to sites of infection
- c. initiate acute-phase responses
- d. have endogenous pyrogen properties
- e. all of the above

46. The alternative pathway of complement activation is characterized by all of the following except:

- a. activation of complement components beyond C3 in the cascade
- b. participation of properdin
- c. generation of anaphylatoxins
- d. activation of C4
- e. regulation by factor H

47. DAF regulates the complement system to prevent complement-mediated lysis of cells. This involves:

- a. dissociation of the C3 convertase complex
- b. blocking the binding of the C3 convertase to the surface of bacterial cells
- c. inhibiting the membrane attack complex from binding to bacterial membranes
- d. acting as a cofactor for the cleavage of C3b
- e. causing dissociation of C5 convertase

48. The following activate(s) the alternative pathway of complement:

- a. lipopolysaccharides
- b. some viruses and virus-infected cells
- c. fungal and yeast cell walls
- d. many strains of Gram-positive bacteria
- e. all of the above

49. A patient with active rheumatoid arthritis feels systemically ill with low-grade fever, malaise, morning stiffness, and fatigue. The protein(s) or cytokine(s) most likely to be responsible for these symptoms are:

- a. rheumatoid factor
- b. TNF and IL-1
- c. IL-4 and IL-10
- d. complement components 1-9
- e. gamma globulin

50. When IL-2 is secreted by antigen-specific T cells activated due to presentation of antigen by APCs, what happens to naive antigen-nonspecific T cells in the vicinity?

- a. They proliferate due to their exposure to IL-2.
- b. They often undergo apoptosis.
- c. They begin to express IL-2 receptors.
- d. They secrete cytokines associated with their Th phenotype.
- e. Nothing happens.

51. Which of the following cytokines plays a role in terminating inflammatory responses?

- a. IL-2
- b. IL-4
- c. TGF- β
- d. IFN- α
- e. IL-3

52. Assuming there were no other compensatory mechanisms to replace IL-8 function, which of the following would be preserved as a functional activity in an IL-8 knockout mouse strain?

- a. activation of neutrophils
- b. attraction of neutrophils to sites of tissue damage
- c. wound healing
- d. extravasation of neutrophils
- e. reduction of cytokine production by TH 1 cells

53. A patient is admitted with multiple bacterial infections and is found to have a complete absence of C3. Which complement-mediated function would remain intact in such a patient?

- a. lysis of bacteria
- b. opsonization of bacteria
- c. generation of anaphylatoxins
- d. generation of neutrophil chemotactic factors
- e. none of the above

54. Complement is required for:

- a. lysis of erythrocytes by the enzyme lecithinase
- b. NK-mediated lysis of tumor cells
- c. phagocytosis
- d. antibody-mediated lysis of bacteria
- e. all of the above

55. Which of the following is associated with the development of SLE-like symptoms?

- a. deficiencies in C1, C4, or C2
- b. deficiencies in C5, C6, or C7
- c. deficiencies in the late components of complement
- d. increases in the serum C3 level
- e. increases in the levels of C1, C4, or C2

56. Molecular mimicry with human myocardial myosin is responsible for myocarditis following infection with and in type 1 diabetes following infection with.....?

- a. Coxsackievirus/ Group A streptococci
- b. Group A streptococci/ coxsackievirus
- c. EBV/CMV d. CMV/EBV
- e. Flu virus/Rubella

57. Which of the following sentence is false regarding the alternative complement pathway?

- a. C3b fragments initiates the pathway
- b. It has the same terminal sequence of events as the classic pathway
- c. triggered by infectious agents in the absence of antibody
- d. activated by mannose-binding lectin
- e. It does not require C1, C2, or C4

58. The following describes COVID RNA vaccine EXCEPT?

- a. a codon-optimized gene of the full-length spike protein, including the signal peptide
- b. cytidine residues are replaced by methyl-3-pseudocytidine
- c. leads to strong antigen-specific humoral and cellular immune responses
- d. a 5'cap (7-methylguanosine) and PolyA tail

59. The following describes nonreplicating viral vaccines EXCEPT?

- a. Each viral particle used as a vector is capable of infecting many cells
- b. Adenoviruses are frequently used as viral vectors
- c. To achieve a strong immune response, more vaccine doses are required
- d. To generate replication-defective vectors, two early genes of adenovirus are replaced by a transgene

60. The important function of the MAC complex formed by complement components is?

- a. To release histamine from mast cells
- b. To inhibit immune complex formation
- c. To enhance antibody production
- d. To opsonize viruses
- e. To perforate bacterial cell membranes

61. The following is false about the programmed cell death protein (PD1)?

- a. Binds to PDL expressed on most tumor cells
- b. Transmit inhibitory signals
- c. competes with CD28 for B7 binding on APCs
- d. expressed by activated T cells

62. The following describes Human papiloma virus (HPV) vaccine EXCEPT:

- a. recombinant DNA vaccine
- b. a major capsid viral protein'
- c. directed against hepatitis B virus
- d. protective against cervical cancer
- e. given to girls of 11-12 years old

63. BCG vaccine is used to protect against while hip against bacteria.

- a. Tuberculosis/ Hemophilus influenza
- b. Hepatitis/ Pertussis
- c. Hepatitis/ Tuberculosis
- d. Hemophilus influenza/ tuberculosis
- e. Tuberculosis/ Hepatitis B

64. Which of the following is NOT a cause of secondary immunodeficiency
Lack of T cells due to mutations which arrest development of the thymic epithelium

65. HIV initially infects cells expressing CCR5

66. Which of the following is NOT a likely reason for the reduction in CD4 T cell numbers characteristic of AIDS
Over production of regulatory T cells

67. Vaccination was invented

by Jenner

68. Vaccines against viruses are

usually Either live-attenuated or killed

69. From this list, the most effective vaccine is

against Tetanus

70. Complement is important in

eliminating Gonorrhoea

71. Activated fragments of C5 can lead to all of the following

except: attachment of lymphocytes to macrophages

72. Antihistamines

block H1 receptors and inhibit smooth muscle contraction, dilatation of small blood vessels, and mucus production

73. Which of the following remains the gold standard for asthma management?

corticosteroids

74. Rejection of a tumor may involve which of the following?

a. T-cell-mediated cytotoxicity

b. ADCC

c. complement-dependent cytotoxicity

d. destruction of tumor cells by phagocytic cells

e. all are correct

75. Which of the following best defines

immunotoxins? toxins coupled to antigen-specific

immunoglobulins

76. Hypersensitivity to penicillin and hypersensitivity to poison oak are

both initiated by haptens

77. Neutrophils are attracted to an infected area

by: phagocytosis of IgE-coated bacteria

78. Complement fixation refers to:

the binding of complement components by antigen-antibody complexes.

79. Natural killer cells are:

able to kill virus-infected cells without prior sensitization

80. A child disturbs a wasp nest, is stung repeatedly, and goes into shock within minutes, manifesting respiratory failure and vascular collapse. This is MOST likely to be due to:

systemic anaphylaxis.

81. The membrane IgM and IgD on the surface of an individual B cell are identical except for their CH regions

82. A delayed hypersensitivity reaction is characterized by: an infiltrate composed of helper T cells and macrophages

83. When immune complexes from the serum are deposited on glomerular basement membrane, damage to the membrane is caused mainly by: enzymes released by polymorphonuclear cells

84. The class of immunoglobulin present in highest concentration in the blood of a human newborn is:

IgG

85. Cytotoxic T cells induced by infection with virus A will kill target cells: infected by virus A and identical at class I MHC loci of the cytotoxic T cells

86. Graft and tumor rejection are mediated primarily by: cytotoxic T cells.

87. The role of the macrophage during an antibody response is to: process antigen and present it